

REFERENCES: READ Reactor Theory Manual Chapter 2, Sections §2.1-§2.3
Delayed Neutron Lecture #6 PPT Slides
PLEASE SHOW ALL WORK!

1. If $\ell_p = 3 \times 10^{-5}$ sec, $\ell_d = 12.7$ sec, $\beta = 0.0073$, and $\rho = +200$ pcm, calculate P_f (Watts) at $t = 10$ sec if $P_i = 100$ Watts. **Reference Eq. 2.1 and 2.3 from NE235 Theory Manual**

2. Referencing *Fig. 2.2 – Neutron Multiplication Cycle* from Lecture #6:
 - a. If we adopt the “storage” mentality as a description of delayed neutron fraction, what is the K_{eff} of the cycle if the ratio of neutrons “placed into storage” (β) equals the release rate of delayed neutrons from storage ($\Sigma \lambda_i C_i$)?

 - b. Which of the six groups (C_i) of delayed neutron precursors takes the longest to reach saturation activity? What is the average half life of this group?

3. As a ‘Light Water Reactor’ (LWR) core like the PULSTAR moves from ‘Beginning of Life’ (BOL) to ‘End of Life’ (EOL) (*mark the following statements True or False*):

<i>T/F?</i>	<i>Statement</i>
	... the amount of ^{235}U in the LWR fuel decreases.
	... the amount of ^{239}Pu in the LWR fuel decreases.
	... the delayed neutron fraction β decreases.
	... the Startup Rate (DPM) increases for a given net positive reactivity (REF FIG 2.4).
	... the reactor power will change more slowly for a given net positive reactivity.

CONTROL ROD CALIBRATION LAB QUESTIONS – REFERENCE EXCEL DATA FILE FROM LAB:

1. Which PULSTAR control rod did we calibrate and which rods did we balance it against to re-establish criticality at each step?
2. Which nuclear instrument channel (e.g. Safety, Linear, Log-N or NFM) on the console did we use to measure the doubling time for each pull?
3. Looking at the excel data file for the lab:
 - a) What was the maximum differential rod worth measured (pcm/in)? At what rod height (in)?
 - b) What was the total sum integral rod worth (pcm) we measured after the last rod pull?
 - c) Why did the measured differential worth increase initially and then start to decrease?